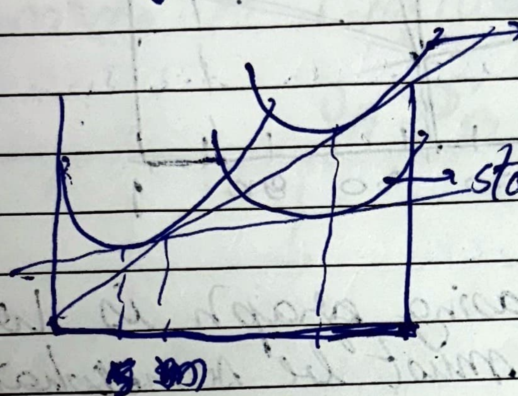
b) → arise of divacancies instead of vacancies changes activation energy

→ Different orientations lead to different activation energies in single crystal

d) activity ≈ chemical potential

c)



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* Diffusion rates non-uniform and slow in amorphous solids.

→ Czochralski Method:-
↳ growth of single crystalline silicon ingot.

→ Ibuprofen:- different morphological crystal shapes.
← based on crystal of seed.

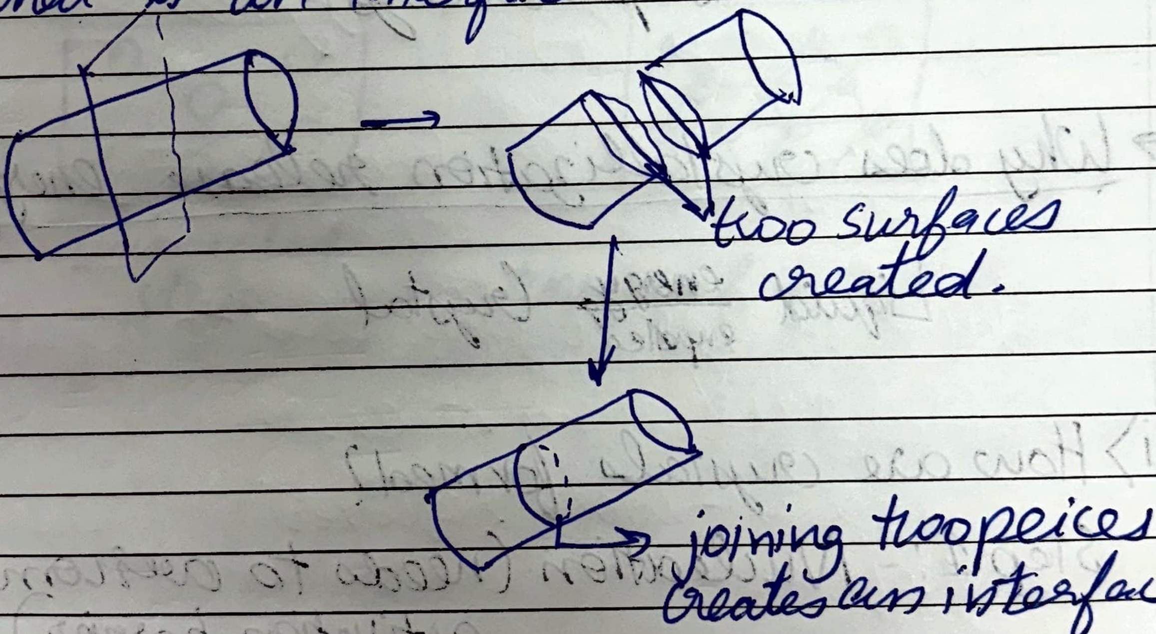
→ Dendritic morphology ~~in~~ during rapid cooling/quenching in alloy systems
→ eg:- Austenite dendrites in steel.

Environment:- Medium of crystal growth
- Characteristics of crystal:- Energy.

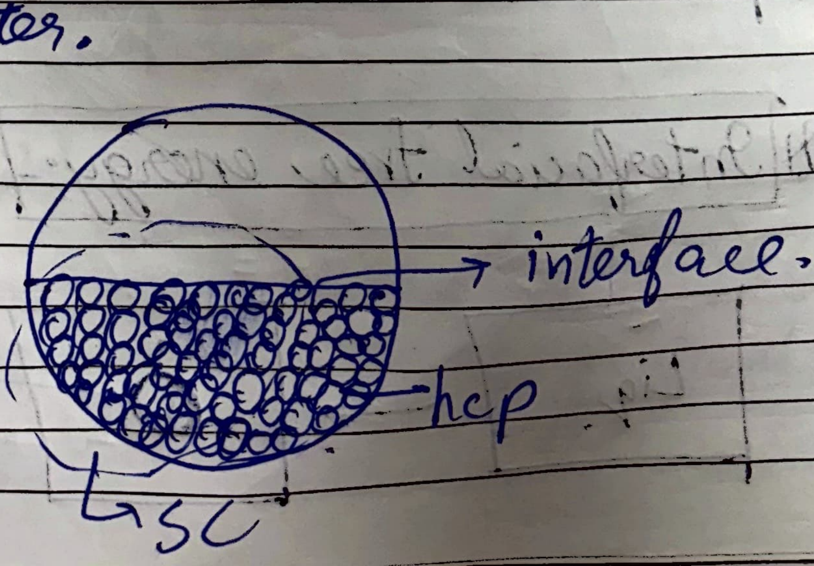
* Equilibrium shape:-
  ↳ shape of very first crystal dictates final growth shape.
→ The final properties depend on growth shape.

→ Surface energy or interfacial energy decides the shape of crystal

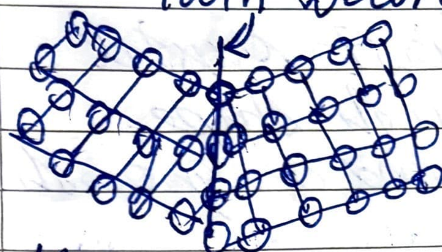
* What is an interface:-



* Boundary b/w two different phases of solid matter.



Homophase interface:-
- Grain boundary
- Twin boundary : forms in single phase



* boundary b/w diff orientations / mirror

Heterophase interface:-
- Interphase interface.

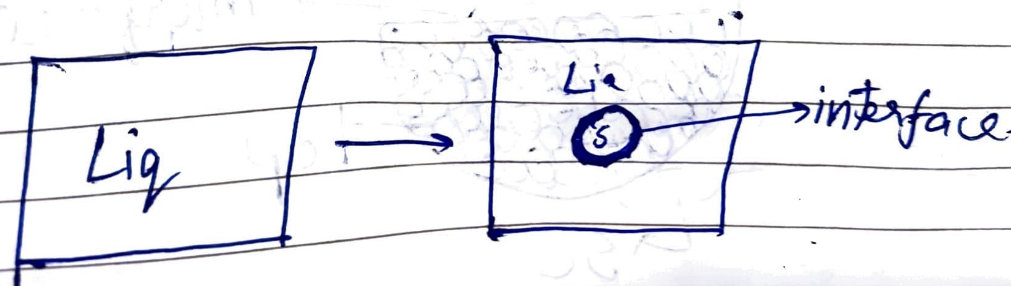
* Ni-based superalloys:-

→ Why does crystallization release energy?

Liquid $\xrightarrow[\text{expelled}]{\text{energy}}$ Crystal

i) How are crystals formed?
Step 1:- Nucleation (needs to overcome activation barrier)
Step 2:- Growth.

$\boxed{\text{Interfacial free energy:-}}$

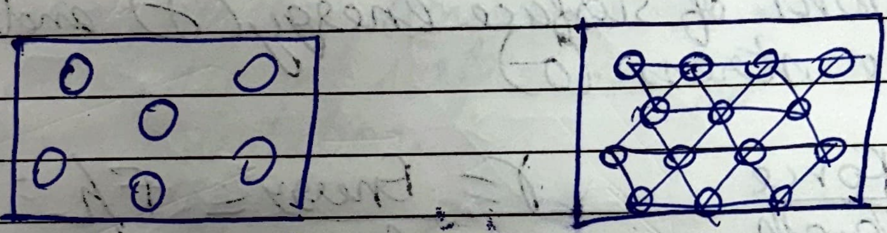


Work done to create a unit volume in 3D : $\underline{dW = PdV}$

→ Work done to create a unit surface in 2D :- $\underline{dw = \gamma dA}$

→ Interfacial energies are always additive. (must be added)

→ formation of interface increases energy.

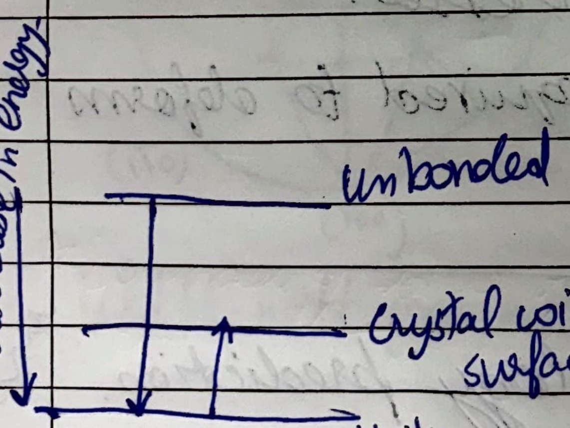


unbonded state.

→ Energy of a crystal with free surface/unit vol

$= [-(\text{no. of bulk atoms}) \times 6E_b]$

$- [-(\text{no. of surface atoms}) \times E_b]$



(due to missing bonds)

→ Energy of a crystal with a free surface/unit volume

$= [-(\text{no. of total atoms}) \times 6E_b]$

$+ [(\text{no. of surface atoms}) \times 2E_b]$

★ Hence nanomaterials have large energy due to huge no. of interfaces and are rather unstable.

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→ Atoms on the surface have a lower coordination no. as compared.

☆ It costs energy to put an atom on the surface as compared to the bulk → origin of Surface energy.

→ The surface wants to minize its area (wants to shrink)

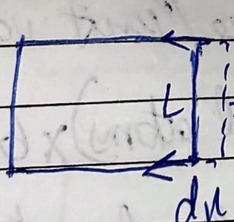
Dimensions of surface energy (γ) and surface stress :- (σ)

$\sigma = \frac{\text{Force}}{\text{length}}$, $\gamma = \frac{\text{Energy}}{\text{Area}} = F/L = N/m$

→ Equivalent for liquids due to capability to withstand shear stresses.

→ Surface stress :- work required to deform the surface.

Particle / Crystal morphology prediction.



change in area.
$\Rightarrow dA = L dx$

$W = F \times s = (2\gamma L) \times dx = 2\gamma L dx$

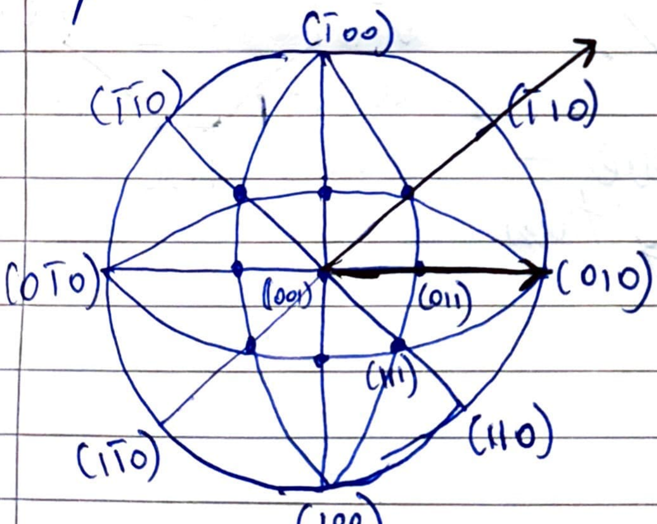
$G = G_0 + A\gamma$

$dG = -SdT + VdP + \sum \mu_i dn_i + \gamma dA$

$\boxed{\gamma = \left(\frac{\partial G}{\partial A}\right)_{T,P,n_i}}$ minimizing area to ~~m~~ have minimal surface energy.

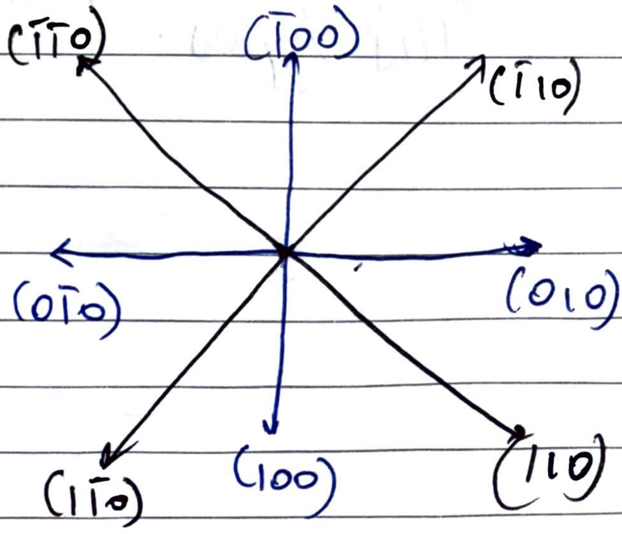
Concept of Wulff's plot:

→ representative construction to show the dependence of surface energy on the crystal plane orientation.



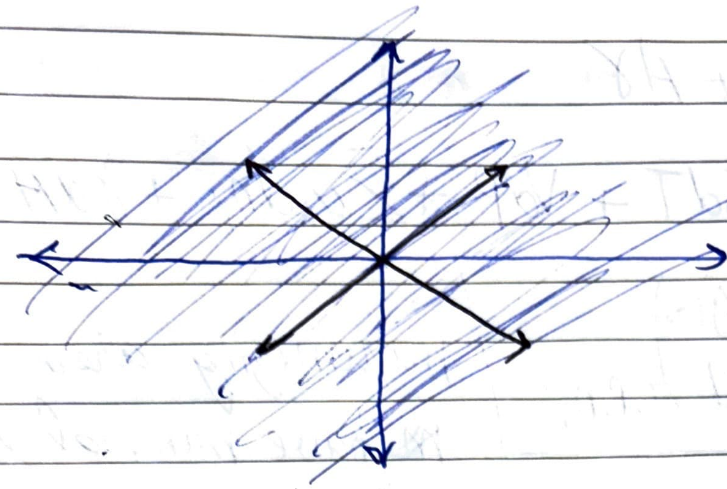
Wolff-nett projections.

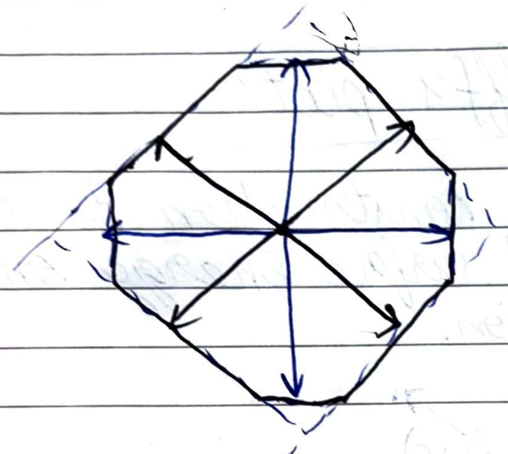
$\gamma_{010} = 0.5\ J/m^2$



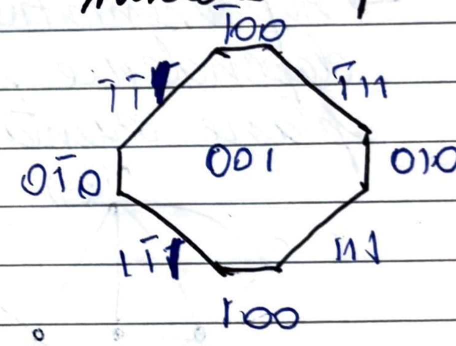
→ Surface energy drawn for lowest surface energy in particular direction.

★ If the.



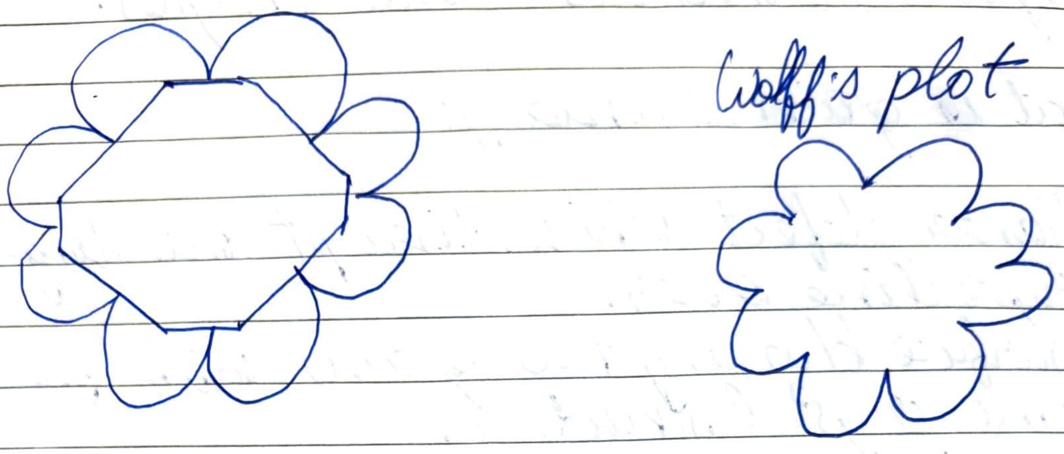


equilibrium nucleus shape.



{100} type planes have looser atomic packing than {111} types.

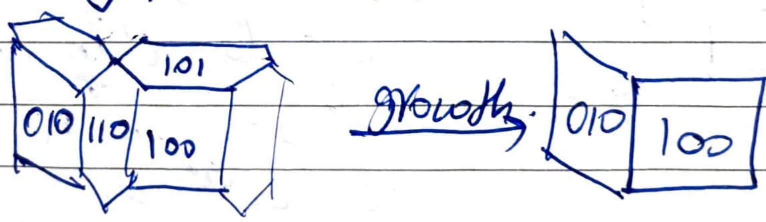
→ Planes whose energies lie at the cusps of Wulff plot have lower energies.



Role of temperature:-

→ Temperature must increase for growth of crystal
→ Temp must ↓se for nucleation probability.

→ Plane with less surface energy and less densely packed shall grow faster.
→ Densely packed planes shall vanish.



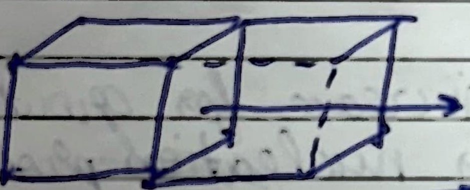
Grain boundary:-

→ Nucleation occurs simultaneously at different locations.
– Random nucleation takes place (both orientation and location is random)
→ boundaries created during impingement.

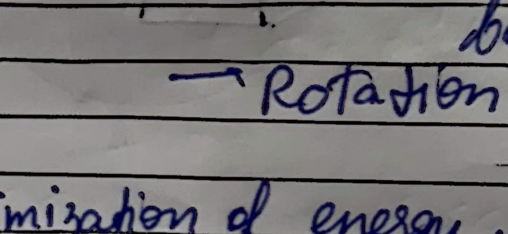
- Grain boundaries are metastable.
- Single crystal is stable.
- Grain boundaries have energies.

What is grain boundary:-

↳ Planar defect which disrupt the long-range crystalline order.
- Surface b/w any two grains with the same crystal structure.
- Grains are related to one another by a rotation axis.



perfectly aligned.
- cannot be distinguished
- boundary not exist.



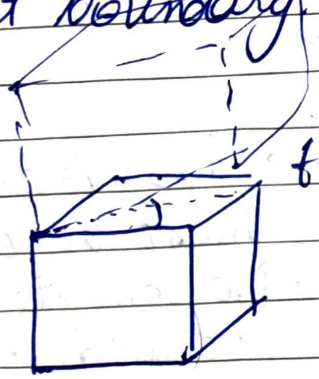
not aligned
angular difference creates diff. class of grain boundaries.

- Rotation axis (reference) is same.

⇒ Minimization of energy, atoms will try maximizing their registery.

Types of grain boundary:-

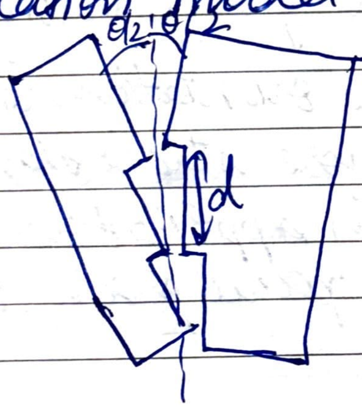
1) Tilt boundary:-



tilt → leads to edge dislocations.

2) Twist boundary.

Dislocation model:-



→ When the misorientation b/w any two grains is small, the boundary formed can be considered as an array of dislocation.

Energy of grain boundary:

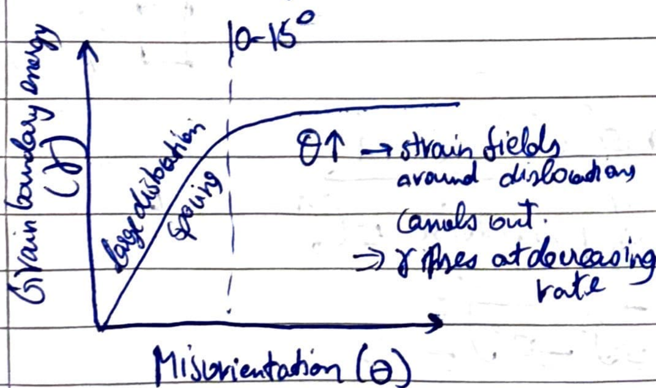
↳ energy of the dislocations within unit area of boundary.

$$\sin\left(\frac{\theta}{2}\right) = \frac{b}{2D}$$

$$\theta = \frac{b}{D}$$

$$(GBE)\ \gamma \propto \frac{1}{D}$$

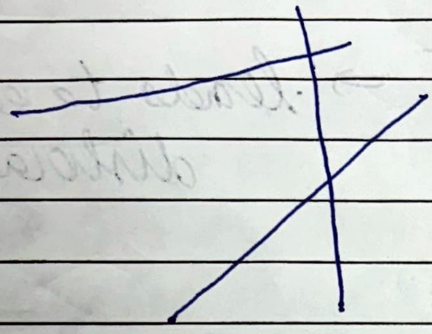
$$\therefore \gamma \propto \theta$$



θ>15° → dislocation spacing too small to distinguish.

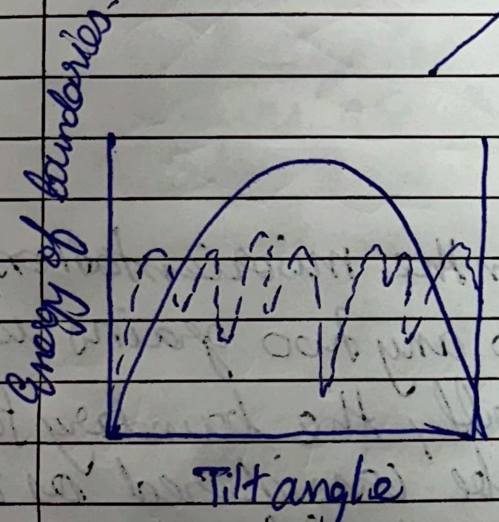
Low angle grain boundary : $\theta < 15^\circ$

High angle grain boundary $\theta > 15^\circ$



→ lines drawn must be consistent

→ No concrete answer obtained depends on orientation chosen.



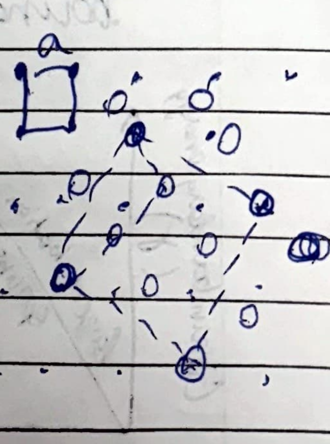
Special grain boundaries

↳ Some orientations (troughs) are more stable than expected

→ certain applications require these special grain boundaries.

Coincident site lattice (CSL) boundaries

→ lattice pts coincides after certain amt of tilt. These lattice pts are called coincident site lattice



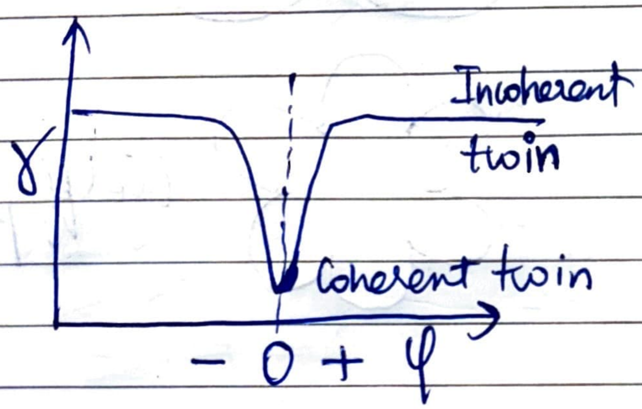
$$\Sigma CSL = \frac{\text{area of CSL cell}}{\text{area of primitive cell}}$$

$$= \frac{\left(\sqrt{(2a)^2 + a^2}\right)^2}{a^2} = 5$$

Twin boundaries:-

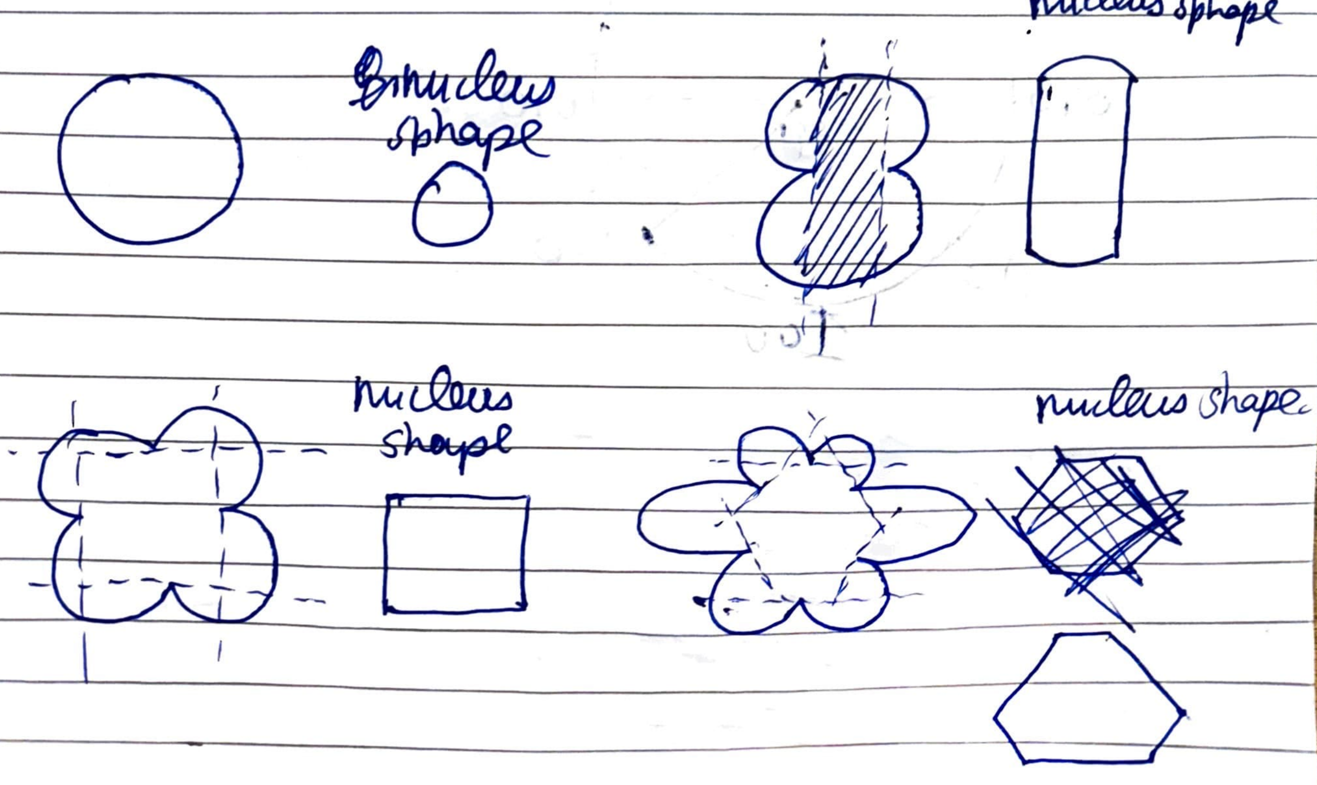
coherent twin boundaries:-

incoherent twin boundary:-
↳ mismatch exists

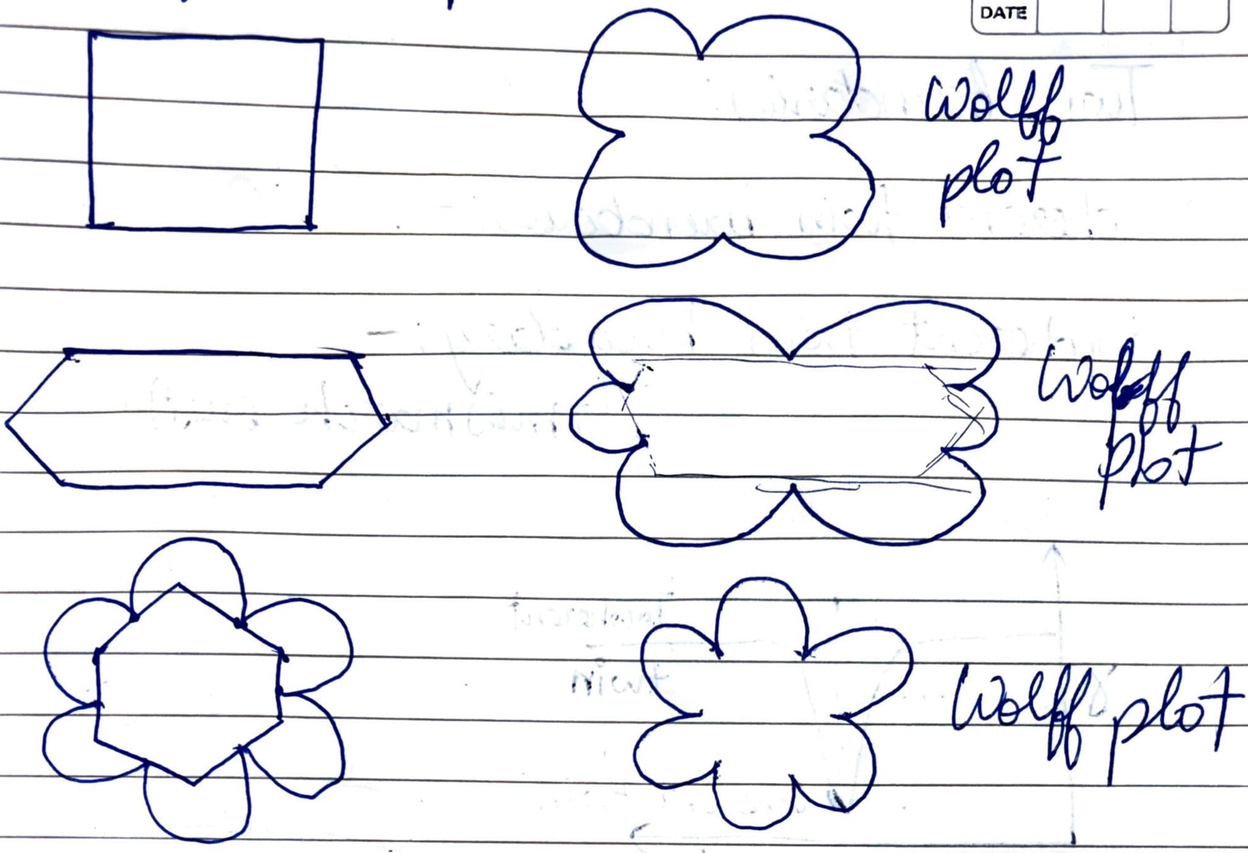


Tutorial

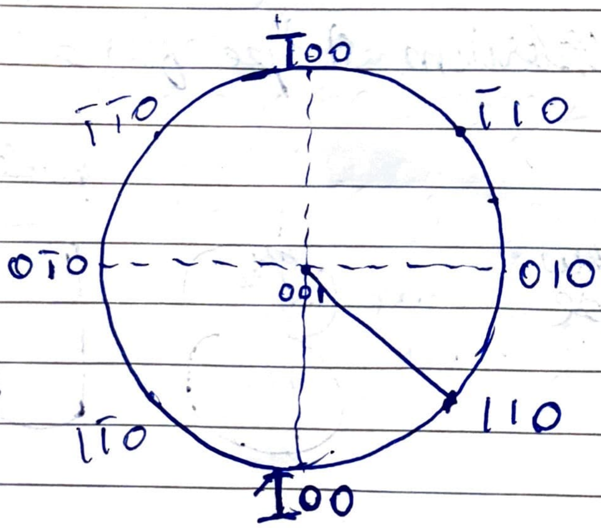
→ Corresponding equilibrium shape for a Wolff plot.



Equilibrium shape



Sterographic Projections :-



Cold Work:-

→ Plastic deformation of a material in the temperature range of $(0.3-0.5)T_m$.

→ Homologous temp of a phase/composition: $\dfrac{\text{Temp of treatment}}{\text{Melting temp}}$

Cold work
- → ↑ses pt defect density
- ↑ses fraction of dislocations. (increases interaction b/w dislocations thereby increasing strength and hardness)
- → ↑ses dislocation density

→ Cold working followed by **Annealing**

↳ Internal stresses need to be reduced which is achieved by annealing.

→ Cold working renders material in metastable state with internal strains

Cold Work → Anneal (softening of material)
- $\xrightarrow{\text{Low T}}$ $\boxed{\text{Recovery}}$
- $\xrightarrow{\text{High T}}$ $\boxed{\text{Recrystallization}}$

⇓

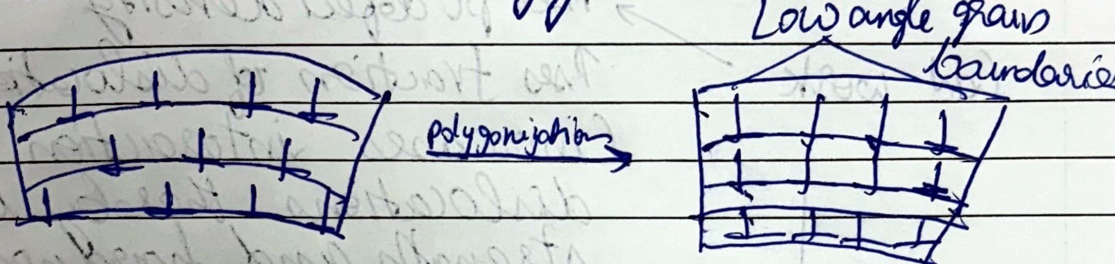
some deformation is possible.

Annealed material $\xrightarrow[\text{work}]{\text{cold}}$ Stronger material

$\rho_{dislocation} \sim (10^5 - 10^9)$ $\rho_{dislocation} \sim (10^{12} - 10^{14})$

Recovery:-

→ configuration of defects rearranges to acheive lower energy.



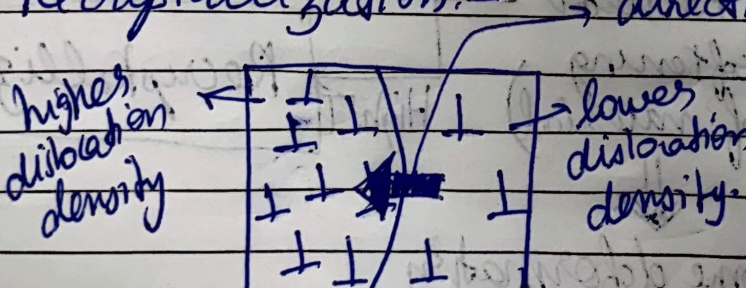
→ extra plane of atoms:- tensile stress
missing plane of atoms:- compressive stress

→ Generation of low angle grain boundaries during recovery

→ Structural dislocations (dislocations generated during plastic deformation)

— These are statistically stored.

Recrystallization:-



→ reduction of stored ~~str~~ strain energy

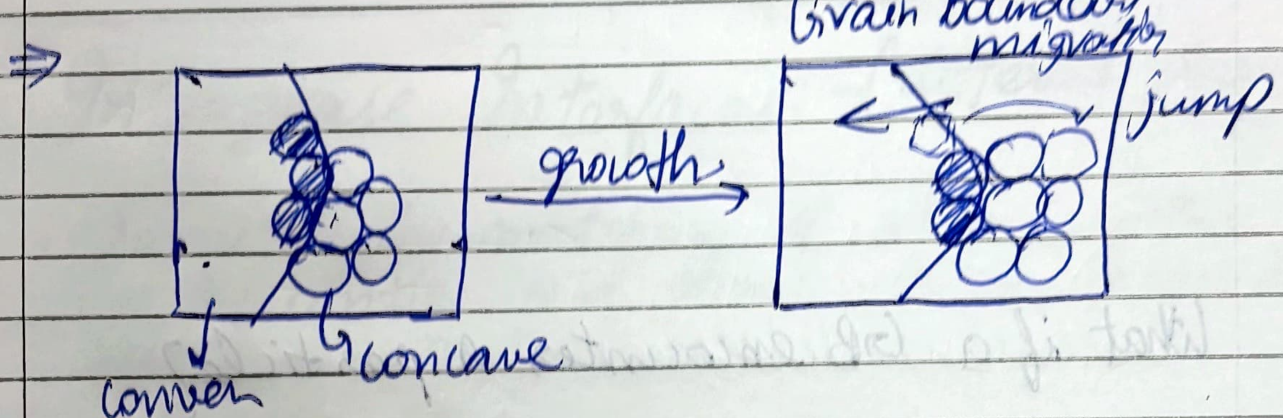
⇒ driving force.

Grain Growth:-

→ lowering of grain boundary energy is the driving force.

→ growth of larger grains at the expense of smaller ones, leading to the increase in the average grain size.

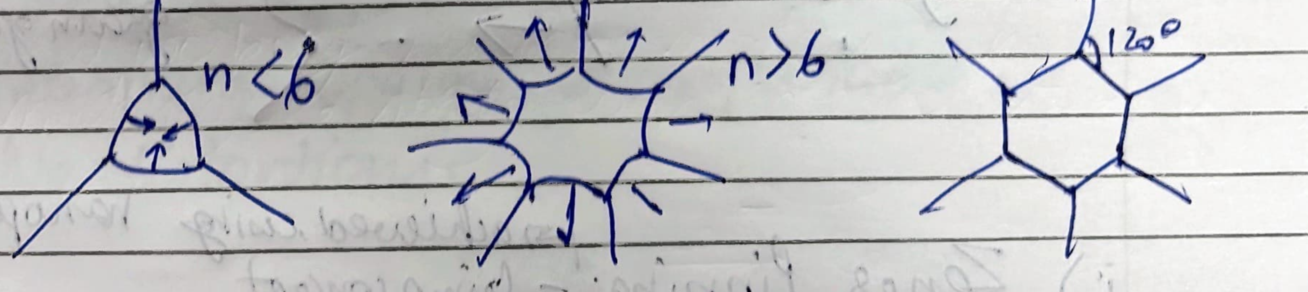
⇒



★ Cells with $n < 6$:- collapse

Cells with $n > 6$:- expand

Cells with $n = 6$:- neither collapse/expand.

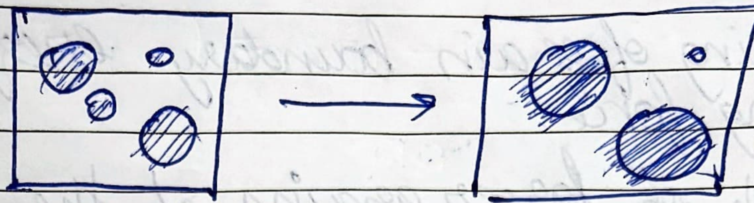


★ Increase in average grain size

Decrease in number of grains.

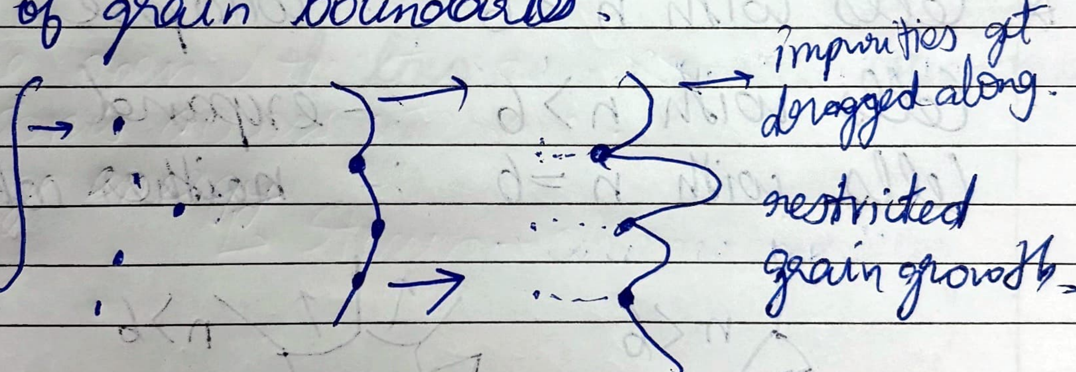
Decrease in GB fraction.

Ostwald ripening:-

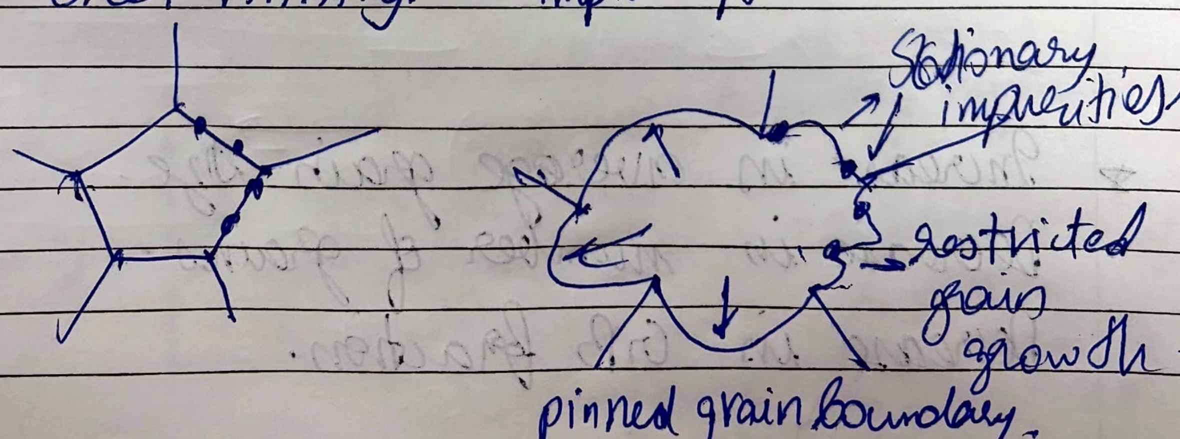


What if a GB encounter a particle?

i) Drag of grain boundaries.



ii) Zener Pinning:- ↳ imp. concept. → achieved using nanoparticles



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$$\boxed{D^2 - D_0^2 = A_0 e^{\left(\frac{-Q}{RT}\right)t}}$$

activation energy to move boundary.

$[t - t_0]$ reference time.

A_0: mobility of grain boundaries [diffusivity]

D_0: diffusity at time t_0

Interphase Interfaces:- [different phases]

→ Based on the matching of atomic sites (or the continuity of atomic planes) across the interface.

i) Coherent interfaces:-

↳ Two crystals match perfectly at the interfacial plane.

→ interplanar spacing is comparable.

→ No distortions.

ii) Strained coherent interfaces.

→ coherent interface with small lattice match.

→ coherency stresses develop in the adjoining crystals.

→ Energy of interface → chemical energy, coherency strain energy.

3) Semi-coherent interface:

$$\text{misfit } (\delta) = \frac{d_\beta - d_\alpha}{d_\alpha}$$

$$b = \text{avg spacing} = \frac{d_\beta + d_\alpha}{2}$$

(burgers vector of dislocation)

4) Incoherent interfaces:-

↳ Inter-atomic distance >25%

→ more open structures, ~~ex~~ lot of movement of atoms across interface.

→ large energy

↳ Strain + chemical energy.